Name: _____

WORKSHEET

Permutations with repetitions

1 Calculate the following:

a $\frac{9!}{3!}$

b $\frac{10!}{4! \times 3!}$

c $\frac{16!}{6! \times 5!}$

d $\frac{12!}{5! \times 6!}$

e $\frac{10!}{2! \times 2! \times 3!}$

f $\frac{12!}{2! \times 3! \times 4!}$

2 For each of the following, write the number of possible anagrams in the form $\frac{n!}{a!}$

a HAPPY

b AMAZING

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c BUBBLE

d HONOURABLE

e REPEATED

f BEEKEEPER

3 Calculate the values of the permutations in question **2**.

a

b

c

d

e

f

4 For each of the following, write the number of possible anagrams in the form $\frac{n!}{a!b!c!...}$.

a CALCULATOR

b PARALLEL

c TRIGONOMETRIC

d MISSISSIPPI

e GRAVITATIONAL

f DEFENCELESSNESS

5 Calculate the values of the permutations in question **4**.

a

b

c

d

e

f

6 How many anagrams are there of each of the following words?

a ELEVATION

b PAPOOSE

c DEMONSTRATION

d CHEERLESSNESS

e UNSUBSTANTIATED

f UNSUCCESSFULLY

7 How many arrangements exist of:

a eight objects if three of them are identical?

b twelve objects if two pairs of them are identical?

c thirteen objects if a pair and a trio are identical?

d fifteen objects if three trios are identical?

e ten objects if a trio and two pairs are identical?

f twenty objects if a pair, two trios and a quadruple are identical?

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Answers

1 **a** 60 480

b 25 200

c 242 161 920

d 5544

e 453 600

f 1 663 200

2 **a** $\frac{5!}{2!}$

b $\frac{7!}{2!}$

c $\frac{6!}{3!}$

d $\frac{10!}{2!}$

e $\frac{8!}{3!}$

f $\frac{10!}{5!}$

3 **a** 60

b 2 520

c 120

d 1 814 400

e 6 720

f 30 240

4 **a** $\frac{10!}{2! \times 2! \times 2!}$

b $\frac{12!}{2! \times 3! \times 4!}$

c $\frac{8!}{2! \times 3!}$

d $\frac{13!}{4! \times 4! \times 2!}$

e $\frac{13!}{3! \times 2! \times 2!}$

f $\frac{15!}{5! \times 2! \times 4!}$

5 **a** 453 600

b 1 663 200

c 3 360

d 5 405 400

e 259 459 200

f 227 026 800

6 **a** $\frac{9!}{2!} = 181\,440$

b $\frac{7!}{2! \times 2!} = 1\,260$

c $\frac{13!}{2! \times 2! \times 2!} = 778\,377\,600$

d $\frac{13!}{4! \times 4!} = 10\,810\,800$

e $\frac{15!}{2! \times 2! \times 2! \times 3! \times 2!} = 13\,621\,608\,000$

f $\frac{14!}{3! \times 3! \times 2!} = 1\,210\,809\,600$

7 **a** $\frac{8!}{3!} = 6\,720$

b $\frac{12!}{2! \times 2!} = 119\,750\,400$

c $\frac{13!}{2! \times 3!} = 518\,918\,400$

d $\frac{15!}{3! \times 3! \times 3!} = 6\,054\,048\,000$

e $\frac{10!}{3! \times 2! \times 2!} = 151\,200$

f $\frac{20!}{2! \times 3! \times 3! \times 4!} = 1\,407\,929\,402\,880\,000$

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